



AI Research Group

CENTRE FOR ADVANCED STUDIES

Dr. A.P.J. Abdul Kalam Technical University
Lucknow



Research Seminar Series

Seminar -7

Brain-Computer Interaction using Artificial Intelligence

DATE & TIME:
17th February 2021,
4:00 PM.

PLATFORM & WEB-LINK

Google Meet :
<https://meet.google.com/waj-ejvk-jou>



SPEAKER

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ABSTRACT

Brain-Computer Interface (BCI) technology is rapidly emerging with an aim to directly establish a pathway connecting the human brain and the computer. Computer-brain interfaces are devised to repair human cognitive functions, restore sensory motor functions, transfer sensory information and trigger the neuronal activity in the brain by means of unnatural electrical signals. BCIs have been studied with two prime goals: a) to enable the communication of disabled population, having any kind of traumatic amputation of limb, motor dysfunctionalities, or any sensory impairment, with the outside world, and b) to promote the detection and diagnosis of mental disorders such as depression, Alzheimer's disease, Parkinson's, Epilepsy and other types of seizure disorders, using brain signals. For a BCI, the neural activities could be picked up by electrophysiological recordings from the brain in a non-invasive way using an electroencephalogram (EEG). In recent years, various machine learning and deep learning techniques have been in practise, sometimes in amalgamation with rehabilitation robotics with a focus on motor imagery (MI) for restoration of motor functions and brain-related disorders. However, these methods should have high accuracy, automatic and highly robust for real time applications. This webinar will emphasise on the basic principles and mechanisms related to mind-machine interface and will throw light on some recent work and projects in the area along with their challenges through some related literature.